

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 227 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Tangents, Normals & Rate of Change	121 PYQs	Tangents & Rates
Very High	Monotonicity, Maxima & Minima	106 PYQs	Optimization

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Tangents, Normals & Rate of Change	$m = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}$, $\text{Tangent: } y - y_1 = m(x - x_1)$, $\text{Normal: } y - y_1 = -\frac{1}{m}(x - x_1)$
Monotonicity, Maxima & Minima	$f'(x) > 0 \implies \text{Strictly Increasing}$, $f'(c) = 0$ & $f''(c) < 0 \implies \text{Local Max}$, $f''(c) > 0 \implies \text{Local Min}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Tangents, Normals & Rate of Change (Very High Yield — 121 PYQs)

Core Principle: Equations of tangent and normal lines, angle of intersection between two curves, and time derivatives as physical rates.

■ **Caution / JEE Pitfall:** Two curves cut orthogonally if and only if the product of their tangent slopes at the intersection point is -1 ($m_1 m_2 = -1$).

■ **Speed Trick:** Length of sub-tangent = $\left|y / \frac{dy}{dx}\right|$; length of sub-normal = $|y \cdot (dy / dx)|$.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Let $f(x)$ be a polynomial of degree 5, and have extrema at $x=1$ and $x=-1$. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^3} = -5$, ..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Let α and β respectively be the maximum and the minimum values of the function $f(\theta) = 4\left(\sin^4\left(\frac{7\pi}{2} - \theta\right) + \sin^4\theta\right)$..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (Numerical): "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that the quadratic equation $f(x)m^2 - 2f'(x)m + f''(x) = 0$ in m , has two distinct real roots for all $x \in \mathbb{R}$..." [Ans: 1]

3.2 Monotonicity, Maxima & Minima (Very High Yield — 106 PYQs)

Core Principle: First and second derivative tests for optimization, global extreme values on closed intervals, and Rolle's/Lagrange's Mean Value Theorems.

■ **Caution / JEE Pitfall:** A critical point occurs where $f'(x) = 0$ OR where $f'(x)$ DOES NOT EXIST (e.g. at $x=0$ for $|x|$).

■ **Speed Trick:** LMVT statement: If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$..." [Ans: B]
- **[2026 Evening Shift] Q#1 (MCQ):** "The least value of $(\cos^2\theta - 6\sin\theta \cos\theta + 3\sin^2\theta + 2)$ is..." [Ans: N/A]
- **[2026 Morning Shift] Q#1 (Numerical):** "Let $(2\alpha, \alpha)$ be the largest interval in which the function $f(t) = \frac{t+1}{t^2}$, $t < 0$, is strictly decreasing. Then the local..." [Ans: 4]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$. Then the minimum value of the function $g(x) = 3 + e^x f(x)$, is:

- (A) $3\left(\frac{e+1}{e}\right)$
- (B) $3\left(\frac{e-1}{e}\right)$ [Correct]
- (C) $(3-e)/e$
- (D) $3e$

Correct Answer: B

Solution Analysis: Given $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0)=3$. We have to find the minimum value of $g(x) = 3 + e^x f(x)$. Step 1: Find the form of $f(x)$. We are given a functional equation. We will use it carefully. Let $\frac{x+y}{3} = t$. Then $x+y=3t$. The relation becomes $f(t) = \frac{f(x)+f(y)}{3}$ whenever $x+y=3t$. We now choose suitable values. Put $x=y=0$. Then $f(0) = \frac{f(0)+f(0)}{3} = \frac{2f(0)}{3}$. So, $3f(0) = 2f(0)$ $\Rightarrow f(0) = 0$. Put $y=0$. Then $f\left(\frac{x}{3}\right) = \frac{f(x)+f(0)}{3} = \frac{f(x)}{3}$. Thus, $f(x) = 3f\left(\frac{x}{3}\right)$. Now differentiate both sides with resp

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Morning Shift (Q#2)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that the quadratic equation $f(x)m^2 - 2f'(x)m + f''(x) = 0$ in m , has two equal roots for every $x \in \mathbb{R}$. If $f(0)=1$, $f'(0)=2$, and (α, β) is the largest interval in which the function $f(\log_e x)$ is increasing, then $\alpha + \beta$ is equal to ____.

Correct Answer: 1

Solution Analysis: For equal roots, discriminant is 0. $4(f''(x))^2 - 4(f(x))(f''(x)) = 0$. Let $y = f(x)$. $(y'')^2 = yy'' \Rightarrow \frac{y''^2}{y} = y'' \Rightarrow \ln|y| = \ln|y''| + \ln K$. $\ln(1) = \ln|2| + \ln(K) \Rightarrow \ln K = -\ln 2$. $\Rightarrow \ln|y| = \ln|y''| - \ln 2$ & $\Rightarrow \ln\left|\frac{y}{y''}\right| = -\ln 2 = \ln\left(\frac{1}{2}\right)$ & $\Rightarrow \frac{y}{y''} = \frac{1}{2}$ & $\Rightarrow 2y = y''$ & $\Rightarrow 2x = \ln|y| + \ln\lambda$ & $0 = \ln|1| + \ln\lambda$ & $\Rightarrow \ln\lambda = 0$ & $\Rightarrow \ln|y| = 2x$ & $y = e^{2x}$